

Problem

In an amplifier circuit, a simple RC high-pass filter is needed to block the dc component while passing the time-varying component. If the desired rolloff frequency is 15 Hz and $C = 10\ \mu\text{F}$, find the value of R .

Step-by-step solution

Step 1 of 1

$$\omega_c = 2\pi f_c = \frac{1}{RL}$$

$$R = \frac{1}{2\pi f_c C} = \frac{1}{(2\pi)(15)(10 \times 10^{-6})} = 1.061\text{ K}\Omega$$